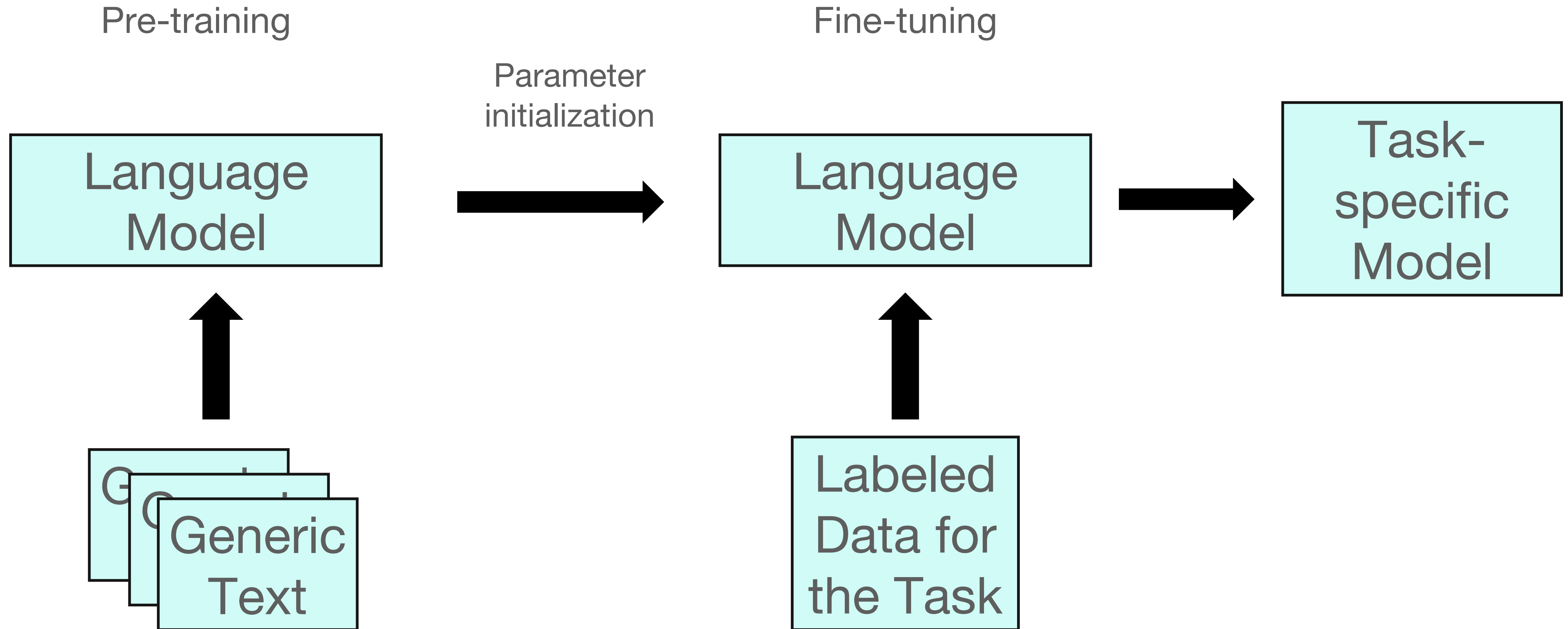


Pre-training

- But for language models all we need is unlabeled text
- We can therefore train language models to perform next word prediction and use the obtained model as initialization for a downstream task
- This stage is known as *pre-training* and the supervised part is called *fine-tuning*
- Pre-training tends to be very demanding computationally, but many fine-tuned models can be used with it

Pre-training and Fine-tuning



Pre-training and Fine-tuning

- For the fine-tuning task, we often replace the linear+softmax layer we used in the LM for decoding, with a task-specific sub-network
- Most common: a classification head – usually a feed forward network and a soft-max over the labels at the end

